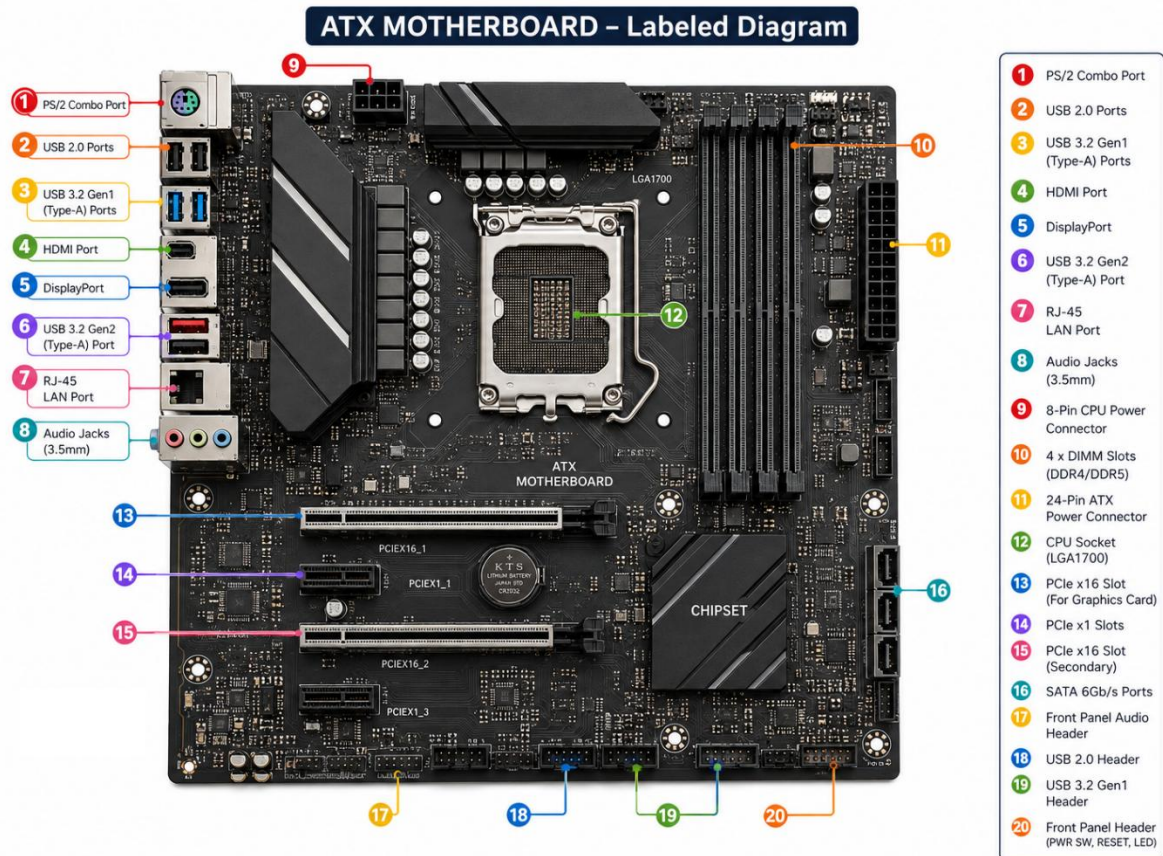


Day 2: Motherboard and CPU Theory

Task : 1 Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



Task 2 : Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

1. LGA1151

- **LGA1151** is a CPU socket developed by Intel and was used in many Intel desktop processors.
- One compatible processor is the **Intel Core i7-9700K**, which is a 9th-generation Intel Core processor with 8 cores and 8 threads.
- It was commonly used in gaming and high-performance desktop computers.

2. AM4

- **AM4** is a popular CPU socket developed by AMD and was used for many AMD Ryzen processors.
- The **AMD Ryzen 5 5600** is a compatible processor with 6 cores and 12 threads, making it suitable for gaming, everyday use, and multitasking.
- AM4 became popular because it supported a wide range of Ryzen CPUs and motherboards.

3. LGA1200

- **LGA1200** is an Intel socket designed for 10th-generation and some 11th-generation Intel desktop processors.
- The **Intel Core i5-10600K** is an example of an LGA1200 processor, featuring 6 cores and 12 threads and offering good performance for gaming and general-purpose computing.

Task 3 : Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU (Central Processing Unit):

- The CPU is the main processor of the computer and is responsible for executing instructions and performing calculations.
- When you open an app like Spotify, the CPU processes the app's instructions, manages its tasks, and helps load and run the application.

Chipset:

- The chipset is a group of controllers on the motherboard that helps different hardware components communicate with each other.
- It manages communication between the CPU, RAM, storage devices, USB devices, audio hardware, and other peripherals.

How They Work Together with Spotify

- When you open **Spotify**, the CPU processes the instructions needed to start the application and loads data from the SSD or HDD through the motherboard's communication system.
- The chipset helps coordinate communication between the CPU, RAM, storage, and audio devices, allowing Spotify to run and play music smoothly.

In simple words: The CPU does the processing, while the chipset helps different hardware components communicate and work together.

Task 4 : Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

